§1.272 — the Cayley representation, and how it relates to the slice A/B

Freyd & Scedrov, §1.272

*The reader's question: "the Cayley representation seems the same as a slice category." Answer: they coincide at the level of **objects** only. §1.272 packages those objects into a functor $A \to \boldsymbol{S}$ by forgetting the slice's triangles and keeping only the right-action of postcomposition — the **regular representation** — which is what embeds A faithfully into $\boldsymbol{S}$.*

§1. §1.272 recap: the Cayley representation

A small category A is automatically a right A-set. The **Cayley representation** is a functor $C : A \to \boldsymbol{S}$ (where $\boldsymbol{S}$ is the category of sets) that sends an object B to the **set**

$$C(B) = \{y : P \to B \mid \text{any source } P\} = \text{all morphisms with target } B.$$

On a morphism $x : A \to B$ it acts by **postcomposition**:

$$C(x) : C(A) \to C(B), \quad y \mapsto yx$$

(diagram order: first $y : P \to A$, then $x : A \to B$, giving $yx : P \to B$).

§2. The coincidence: $C(B) =$ objects of A/B

The slice category A/B has as its **objects** exactly the arrows into B — every morphism $y : P \to B$ (for any source P) is an object of A/B. So

$$C(B) = \text{the set of objects of } A/B.$$

That overlap is real and not accidental.

§3. The difference: category vs bare set

The two gadgets share the same underlying collection of objects, but they are different kinds of things.

- ***A/B* is a category.** Besides the arrows-into-B (its objects) it keeps the **triangles** between them: a morphism $(y : P \to B) \to (y' : P' \to B)$ in A/B is a map $t : P \to P'$ with $ty' = y$ (diagram order: first t, then y'). Those triangles are the point of a slice.
- ***C(B)* is a bare set.** C forgets the triangles and remembers only the set of arrows into B together with how postcomposition $y \mapsto yx$ moves it. That action is the "representation."

So: $C(B) =$ objects of A/B, with the slice's morphisms **forgotten**, repackaged as a $\boldsymbol{S}$-valued functor.

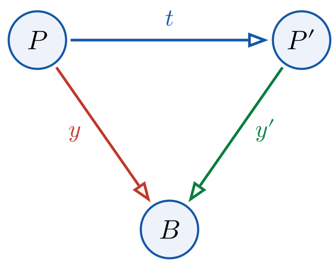

Figure 1: A morphism in A/B from the object $(y : P \to B)$ to the object $(y' : P' \to B)$: a map $t : P \to P'$ making the triangle commute ($ty' = y$, diagram order). **$C(B)$ forgets t entirely** — it keeps only the set $\{y, y', \ldots\}$ of vertices together with the right-action of postcomposition.

§4. Two sharper framings

1. C is the regular representation, not a single representable.

$$C(B) = \coprod_{P \in A} \mathrm{Hom}(P, B)$$

— the coproduct of **all** covariant representables $\mathrm{Hom}(P, -)$ over every object P of A. (Contrast $\mathrm{Hom}(P, -)$, which fixes one source P.) Summing over all sources is exactly "A acting on its own morphisms," i.e. A as a right A-set.

2. The monoid case makes it vivid.

Take A to be one object $*$ (a monoid M). Then:

- $C(*) =$ all morphisms $= M$ itself, a set with right-multiplication $y \mapsto yx$ — the classical **right regular representation**.
- $A/*$ is instead a **category**: its object-set is M, but it also has triangles t with $tx = y$; for a group those make it a contractible groupoid (a torsor).

Same object-set M; different gadget once you look at morphisms.

⋆ Punchline

The objects of A/B **are** the Cayley set $C(B)$. The Cayley representation then **forgets the triangles** and keeps only the right-action, turning that object-set into a functor $A \to \boldsymbol{S}$. That is what embeds A faithfully into $\boldsymbol{S}$ (§1.272: every small category is isomorphic to a concrete category): postcomposition $y \mapsto yx$ on C is injective — two distinct morphisms $x, x' : A \to B$ act differently on some element of $C(A)$ — giving a faithful functor.